

Multiscale Dynamic Feature Reconstruction and Two-Channel Spatial Enhancement for Remote Sensing Multispectral-Panchromatic Image Fusion

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Scientific integrity note. The original design material specifies the proposed architecture but does not provide actual datasets used, train/test splits, hardware specifications, measured quantitative scores, or experimental figures. Therefore, numerical results have not been invented. Section 4 contains a publication-ready experimental protocol and clearly marked [Measured value] fields that must be replaced by real outputs after implementation and testing.

Abstract

The fusion of a low-spatial-resolution multispectral image with a high-spatial-resolution panchromatic image, commonly referred to as pansharpening, is an important technique for obtaining remote sensing imagery with simultaneously enhanced spatial detail and spectral information. However, conventional fusion methods often suffer from spectral distortion, spatial-detail loss, and insufficient adaptation to objects at different scales. Although deep-learning-based methods have considerably advanced pansharpening, directly coupling multispectral and panchromatic features within a single network may introduce undesirable spatial-spectral interference. To address these limitations, this paper proposes a multistage remote sensing image fusion framework based on multiscale dynamic feature reconstruction and two-channel panchromatic spatial enhancement, referred to as MDFTC-Fusion. First, a multiscale feature dynamic interaction network is introduced to reconstruct a high-resolution spectral representation from the low-resolution multispectral input. This network consists of a Multi-Scale Feature Extraction Module, a Dynamic Feature Fusion Module based on self-attention, and a residual-purpose U-shaped neural reconstruction module. Second, a two-channel neural network independently enhances the spatial information of the panchromatic image by combining shallow features extracted through a three-layer convolutional branch with deep contextual information generated by a multiscale branch, followed by four residual blocks. Finally, the reconstructed multispectral RGB image is transformed into the intensity-hue-saturation domain, where the enhanced panchromatic information is incorporated into the intensity component before inverse transformation. The proposed design explicitly separates spectral reconstruction, panchromatic spatial enhancement, and final fusion, thereby reducing the spatial-spectral conflict encountered by direct feature concatenation methods. A comprehensive evaluation framework based on reduced-resolution and full-resolution assessment, comparisons with representative traditional, convolutional, transformer-based, adaptive, and generative approaches, and component-level ablation experiments is established to validate the proposed architecture.

Keywords: remote sensing image fusion; pansharpening; multispectral image; panchromatic image; multiscale feature extraction; self-attention; residual learning; U-shaped network; IHS fusion; deep learning.

1. Introduction

Remote sensing satellites provide indispensable Earth-observation data for urban planning, agricultural monitoring, disaster assessment, environmental analysis, land-cover mapping, forestry, transportation management, and many other scientific and industrial applications. Nevertheless, due to physical limitations in imaging sensors, satellite systems commonly face a trade-off between spatial resolution and spectral resolution. A multispectral sensor acquires information in multiple spectral bands and therefore provides strong discrimination among different surface materials, whereas a panchromatic sensor integrates radiance over a broader spectral interval and usually offers considerably finer spatial detail. Pansharpening addresses this trade-off by combining a low-spatial-resolution multispectral image with a corresponding high-spatial-resolution panchromatic image to produce an image possessing both rich spectral information and fine spatial structures. Modern surveys generally categorize pansharpening approaches into component-substitution methods, multiresolution-analysis methods, model-based optimization approaches, and data-driven deep neural networks [1].

Let the low-resolution multispectral image be denoted by

$$M \in \mathbb{R}^{(h \times w \times C)},$$

where h , w , and C represent spatial height, spatial width, and the number of spectral bands, respectively. The corresponding high-resolution panchromatic image is represented as

$$P \in \mathbb{R}^{(H \times W)},$$

where generally

$$H = r h, \quad W = r w,$$

and r denotes the spatial-resolution ratio, commonly equal to four in many satellite-imaging configurations. The objective is to estimate a high-resolution multispectral image

$$F \in \mathbb{R}^{(H \times W \times C)}$$

that retains the spectral characteristics of M while incorporating the structural and spatial details contained in P .

Traditional component-substitution techniques, including intensity-hue-saturation transformation, principal-component analysis, and Gram-Schmidt-based methods, usually offer low computational complexity and visually sharp outputs. However, replacing a spectral intensity component with panchromatic information can modify inter-band relationships and introduce color or spectral distortion. Multiresolution-analysis approaches inject high-frequency details extracted from the PAN image into the multispectral data, but their performance depends strongly on filtering strategy, decomposition scale, sensor characteristics, and parameter selection. Deep neural networks have consequently become increasingly important because they can learn nonlinear mappings between multispectral and panchromatic observations and the desired high-resolution representation [2].

The first generation of convolutional pansharpening networks demonstrated that relatively shallow CNNs could learn effective image-fusion mappings. PNN adapted a three-layer super-resolution architecture to pansharpening; PanNet incorporated domain knowledge through high-pass processing; MSDCNN introduced multiscale and multidepth feature extraction; DiCNN explicitly learned details to be injected into an upsampled multispectral image; and TFNet adopted separate feature streams for PAN and multispectral data. These developments established convolutional feature learning as a major direction for remote sensing image fusion [2-7].

More recently, residual learning, self-attention, transformer architectures, adaptive convolutions, deep unfolding, generative adversarial networks, diffusion models, and parameter-efficient adaptation have significantly broadened pansharpening research. For example, transformers enable long-range dependence modeling, adaptive convolutions improve content-dependent feature extraction, unsupervised frameworks reduce reliance on artificial reduced-resolution ground truth, and diffusion models provide alternative mechanisms for modeling complex spatial-spectral distributions [8-30].

Despite these advances, three fundamental difficulties remain.

First, the LRMS and PAN observations have intrinsically different information characteristics. The multispectral image is spectrally rich but spatially coarse, whereas the panchromatic image is spatially detailed but spectrally incomplete. Directly concatenating them at the input layer and expecting a single network to simultaneously resolve both modalities may produce competition between spatial-detail injection and spectral preservation.

Second, remote sensing scenes contain structures at substantially different scales. Roads, narrow building boundaries, vehicles, vegetation regions, agricultural fields, rivers, and large urban blocks require receptive fields with different spatial extents. A single convolutional kernel size is insufficient to model this heterogeneous content efficiently. Although stacking many convolutional layers enlarges the effective receptive field, information degradation and insufficient interaction between features of different scales may still occur.

Third, indiscriminate incorporation of PAN details can cause spectral distortion. High-frequency components in a panchromatic image are not always perfectly consistent with every multispectral band because the sensors may have different spectral-response functions, acquisition conditions, noise properties, and spatial alignment errors. Therefore, robust fusion should not simply force all PAN information into the multispectral representation.

Motivated by these challenges, this study proposes MDFTC-Fusion, a structured fusion system that decomposes the pansharpening problem into three complementary stages. Instead of immediately mixing LRMS and PAN signals within one black-box network, the proposed method first reconstructs the multispectral input at high spatial resolution, independently enhances the panchromatic observation, and finally performs intensity-domain fusion.

The first stage employs a multiscale feature dynamic interaction network containing three main submodules: a Multi-Scale Feature Extraction Module (MFEM), a Dynamic Feature Fusion Module (DFFM), and a residual-purpose neural reconstruction module based on a U-shaped encoder-decoder. MFEM extracts information using convolutional operations with different receptive fields. DFFM employs self-attention to adaptively determine the importance of multiscale responses. The residual-purpose module subsequently reconstructs the high-resolution spectral image by combining local details, contextual information, and encoder-decoder skip connections.

The second stage uses a two-channel panchromatic enhancement network. One branch consists of a three-layer basic convolutional module for extracting shallow textures and edges. A parallel multiscale feature-extraction branch captures deeper and more contextual structures. Their representations are combined and refined by four residual blocks. The predicted spatial-detail residual is finally added to the original PAN image, allowing the network to enhance spatial information while preserving the fundamental radiometric structure of the original observation.

The third stage performs image fusion in the IHS color space. The high-resolution reconstructed R, G, and B bands are transformed into intensity, hue, and saturation components. The enhanced PAN image is radiometrically normalized to the intensity component and then fused with that component. Inverse transformation generates the final high-resolution color multispectral result.

The principal contributions of this paper are summarized as follows.

- First, a staged spatial-spectral fusion architecture is introduced to reduce interference between spectral super-resolution and PAN-detail enhancement.
- Second, a multiscale dynamic reconstruction network is developed to recover high-resolution multispectral information through multiscale convolution, self-attention-based adaptive feature fusion, and residual U-shaped reconstruction.
- Third, a two-channel PAN enhancement module is proposed to jointly exploit shallow structural details and deep multiscale information, followed by residual refinement and global image-level residual learning.
- Fourth, the reconstructed multispectral information and enhanced panchromatic information are integrated through a radiometrically controlled IHS fusion process, providing an interpretable final fusion stage.

The remainder of this paper is organized as follows. Section 2 reviews related pansharpening literature. Section 3 presents the proposed MDFTC-Fusion framework in detail. Section 4 defines the implementation, evaluation, comparison, and ablation procedures. Section 5 summarizes the proposed methodology, current limitations, and future directions.

2. Related Work

2.1 Traditional and early deep-learning pansharpening

Pansharpening has been studied extensively through component substitution, multiresolution analysis, variational optimization, sparse representation, Bayesian modeling, and deep learning. Comprehensive reviews have emphasized that no single traditional formulation simultaneously guarantees perfect spatial fidelity, spectral preservation, computational efficiency, robustness to misregistration, and generalization across different satellite sensors. These limitations motivated the transition toward data-driven feature extraction [1].

Masi et al. introduced PNN, one of the earliest CNN architectures explicitly developed for pansharpening. PNN demonstrated that a three-layer CNN inspired by image super-resolution could effectively estimate an HRMS image from multispectral and panchromatic observations [2]. Yang et al. subsequently developed PanNet, incorporating domain-specific prior knowledge by learning primarily in a high-pass domain and using residual connections to preserve spectral content [3]. Yuan et al. proposed MSDCNN, in which multiscale and

multiscale convolutional structures capture image information at several receptive-field sizes [4]. These studies provided strong evidence that multiscale representation and residual learning are particularly suitable for remote sensing imagery.

He et al. formulated pansharpening within a detail-injection CNN framework, directly predicting the multispectral detail information required to sharpen the LRMS observation [5]. Liu et al. proposed TFNet, which employs two separate streams to extract PAN and MS features before feature-level fusion and image reconstruction [6]. This two-stream strategy is particularly relevant because it acknowledges the heterogeneous characteristics of the two sensing modalities rather than treating them as identical input channels. Deng et al. later developed FusionNet, integrating deep residual learning with the classical detail-injection concept and demonstrating the value of connecting physically interpretable fusion principles with learned nonlinear representation [7].

These early deep models substantially improved reconstruction quality but also exposed several challenges. Supervised training commonly relies on Wald-type reduced-resolution protocols because true HRMS ground truth does not generally exist at the original PAN spatial resolution. Consequently, networks trained exclusively at reduced resolution may encounter scale-transfer problems when applied to full-resolution data. Furthermore, standard convolutions use spatially invariant kernels, which may not adapt optimally to diverse land-cover regions and object structures.

2.2 Unsupervised and full-resolution learning

To mitigate the lack of true ground truth, unsupervised and self-supervised pansharpening have become important research directions. Qu et al. developed an unsupervised pansharpening framework based on self-attention, demonstrating that explicit target HRMS images are not always necessary when appropriate spatial and spectral constraints are designed [8]. Ciotola et al. introduced a full-resolution CNN training framework that defines complementary spatial and spectral losses directly at the native image scale, thereby avoiding exclusive dependence on synthetically degraded ground truth [9].

Such approaches address a central limitation of supervised pansharpening: the unavailable true full-resolution reference. Nevertheless, their effectiveness depends heavily on the accuracy of sensor models, loss-function design, local correlation assumptions, registration quality, and the balance between spatial and spectral constraints. A model may achieve strong spectral consistency while producing insufficient spatial sharpening, or inject strong high-frequency information at the cost of color distortion.

The present study follows a different but complementary philosophy. Instead of relying only on a loss function to resolve spatial-spectral competition, it explicitly decomposes the processing flow into high-resolution multispectral reconstruction, PAN enhancement, and final IHS-domain fusion. This decomposition makes the origin of each information component more transparent.

2.3 Transformers, attention, and model-guided architectures

Self-attention and transformers have been introduced into pansharpening to capture global contextual dependencies that may be difficult to model using local convolution alone. Su et al. proposed a transformer-based regression network for remote sensing pansharpening, exploiting attention to coordinate spatial and spectral information [10]. Meng et al. developed a Vision Transformer architecture for pansharpening, further illustrating the usefulness of global information interaction [11].

Model-based deep architectures represent another important direction. Rather than learning an unrestricted input-to-output mapping, these approaches formulate an observation model or optimization process and then unfold iterative computations into neural-network layers. Cao et al., for instance, proposed a model-based deep pansharpening network that derives learnable processing stages from a convolutional sparse-coding formulation [12]. Zhang et al. introduced spatial and spectral extraction combined with adaptive feature fusion to improve the complementary use of modality-specific information [13].

The proposed MDFTC-Fusion framework shares the goal of preserving interpretability but adopts a different structure. The multispectral branch explicitly performs spectral-image super-resolution, the PAN branch explicitly performs spatial enhancement, and the IHS fusion stage explicitly combines their complementary

outputs. Within the multispectral reconstruction network, self-attention is not used as an unrestricted end-to-end transformer backbone; instead, it serves as a dynamic mechanism for weighting features extracted at different scales.

2.4 Recent convolutional, residual, progressive, and flow-based approaches

A substantial number of advanced CNN-based methods appeared in 2023. Peng et al. introduced PSCF-Net, a deeply coupled feedback network designed to improve progressive information interaction [14]. Fang et al. proposed SDRCNN, a single-scale dense residual-connected network emphasizing a favorable balance between reconstruction accuracy and computational efficiency [15]. Liu et al. proposed Mun-GAN, a multiscale unsupervised generative network that addresses pansharpening without requiring conventional paired full-resolution labels [16]. Yang et al. formulated PanFlowNet from a flow-based modeling perspective, emphasizing the ill-posed nature of HRMS reconstruction [17].

Self-supervised learning has also been investigated. He et al. proposed sSRPNet, using spectral super-resolution principles and coarse-scale self-learning to exploit internal multispectral information [18]. MPEFNet introduced multilevel progressive enhancement for remote sensing image fusion [19]. Cross-spectral and cross-spatial nonlocal attention mechanisms have been developed to model longer-range interactions between LRMS and PAN features [20]. CGSNet employed channel grouping and feature shuffling to enhance feature communication efficiently [21].

These methods demonstrate that effective pansharpening benefits from multilevel processing, iterative refinement, residual connections, attention, and separate consideration of spectral and spatial representations. However, increasingly complex end-to-end architectures may make it difficult to determine whether performance gains arise from better multispectral reconstruction, better PAN-detail extraction, or more effective cross-modal interaction. MDFTC-Fusion therefore isolates these functional objectives into dedicated stages.

2.5 Adaptive convolution, diffusion, progressive fusion, and recent architectures

Duan et al. proposed content-adaptive nonlocal convolution and CANNNet, in which regional adaptation and nonlocal self-similarity improve feature extraction beyond conventional spatially invariant convolution [22]. Shu et al. developed the Cross Modulation Transformer, dynamically coordinating PAN and multispectral information while using a hybrid loss incorporating frequency-domain considerations [23]. Zhong et al. introduced SSDiff, a spatial-spectral integrated diffusion model with separate branches and alternating-projection-based fusion [24].

Shen et al. developed USCPNet, an unsupervised pansharpening approach incorporating imaging spectral priors and spatial-spectral compensation [25]. Wen et al. proposed a multistage progressive pansharpening network that gradually increases spatial resolution and uses attention and explicit spatial-detail extraction [26]. These developments highlight the continued importance of staged processing rather than one-step feature mixing.

Wu et al. subsequently introduced PanAdapter, which uses parameter-efficient adaptation and pretrained image-restoration representations to address the relatively small size of pansharpening training datasets [27]. Ding et al. proposed the Cross-Domain Fusion Attention Network, integrating multidomain interaction and spatial multiscale enhancement [28]. Ma et al. developed a deep spatial-spectral fusion transformer that explicitly models task-relevant relationships between spatial and spectral properties [29]. More recent hybrid-attention architectures continue to investigate coordinated spatial-spectral feature interaction [30].

The literature consequently indicates several consistent trends: multiscale extraction is important because remote sensing scenes contain heterogeneous structures; attention mechanisms can adaptively control feature interactions; residual learning helps preserve source information while estimating missing details; and explicit separation of spatial and spectral processing can reduce destructive competition between the modalities. The method proposed in this paper combines these insights in a three-stage framework centered on dynamic multiscale spectral reconstruction, two-channel PAN enhancement, and interpretable IHS-domain fusion.

3. Proposed Method

3.1 Overview of the MDFTC-Fusion architecture

The objective of the proposed system is to generate a high-resolution multispectral image by combining a low-resolution multispectral observation with a high-resolution panchromatic observation. The complete processing flow consists of three stages:

1. High-resolution spectral reconstruction of the LRMS image;
2. Spatial-information enhancement of the PAN image;
3. Fusion of the reconstructed multispectral image and enhanced PAN image in the IHS domain.

For clarity, let

$$M = \{M_1, M_2, \dots, M_C\}$$

denote the LRMS input. For the RGB implementation of the proposed IHS fusion stage, the red, green, and blue bands are written as

$$M_{\text{RGB}} = [M_R, M_G, M_B].$$

The PAN image is represented by P . The complete mapping can be expressed as

$$F = T_{\text{IHS}}^{-1} (G(T_{\text{IHS}}[S_{\text{MS}}(M_{\text{RGB}})], E_{\text{PAN}}(P))),$$

where $S_{\text{MS}}(\cdot)$ denotes the multispectral super-resolution network, $E_{\text{PAN}}(\cdot)$ represents the two-channel PAN enhancement network, T_{IHS} and T_{IHS}^{-1} are forward and inverse IHS transformations, and $G(\cdot)$ is the intensity-fusion operation.

This separation is deliberate. The multispectral reconstruction branch concentrates on recovering a spatially enhanced but spectrally consistent representation. The PAN network focuses exclusively on structural-information enhancement. The last stage performs controlled information integration.

3.2 High-resolution spectral imaging of low-resolution remote sensing multispectral images

3.2.1 Multispectral preprocessing and initial upsampling

Because the spatial dimensions of the LRMS image are smaller than those of the PAN image, an initial resolution-alignment operation is required. For scale ratio r , the LRMS input is upsampled as

$$M^{\text{up}} = U_r(M),$$

where U_r represents interpolation by factor r . According to the architecture described in the source material, $r = 4$ is adopted for the principal implementation.

Although bicubic interpolation can generate M^{up} , interpolation alone cannot reconstruct genuinely missing high-frequency information. It therefore serves only as spatial alignment and an initial estimate.

The proposed multiscale feature dynamic interaction network comprises the following sequence:

$$\text{MFEM} \rightarrow \text{DFFM} \rightarrow \text{Residual U-shaped Reconstruction}.$$

3.2.2 Multi-Scale Feature Extraction Module

Remote sensing scenes include structures with strongly varying scales. Consequently, the Multi-Scale Feature Extraction Module employs parallel transformations with different effective receptive fields.

Given an input feature tensor X , three representative branches may be formulated as

$$\begin{aligned} F_1 &= \text{phi}(W_{(3 \times 3)} * X + b_1), \\ F_2 &= \text{phi}(W_{(5 \times 5)} * X + b_2), \\ F_3 &= \text{phi}(W_{(3 \times 3)}^{(d=2)} * X + b_3), \end{aligned}$$

where $*$ represents convolution, $d = 2$ denotes dilation, and $\text{phi}(\cdot)$ is a nonlinear activation function.

The branches have complementary functions. The smallest receptive field is especially useful for narrow edges and fine textures. The larger kernel captures intermediate contextual information. The dilated convolution provides a larger effective receptive field without excessive parameter growth.

The branch outputs are initially concatenated:

$$F_MS = \text{Concat}[F_1, F_2, F_3].$$

A 1×1 projection then controls the feature dimensionality:

$$F_P = W_{(1 \times 1)} * F_MS.$$

Unlike simple concatenation-based multiscale designs, the proposed method does not assume that all scales have equal importance. The resulting representations are subsequently processed by the Dynamic Feature Fusion Module.

3.2.3 Dynamic Feature Fusion Module

A fixed sum or fixed concatenation of multiscale features cannot adapt to heterogeneous land-cover regions. For example, a dense urban scene may require stronger fine-edge representation, while a broad agricultural area may benefit more from medium- or large-scale contextual information.

The Dynamic Feature Fusion Module therefore employs self-attention to determine input-dependent relationships among multiscale features. For feature tensor F_P , query, key, and value representations are generated:

$$Q = W_Q F_P, \quad K = W_K F_P, \quad V = W_V F_P.$$

The self-attention matrix is

$$A = \text{Softmax}(Q K^T / \sqrt{d_k}),$$

and the dynamically fused representation becomes

$$F_D = A V.$$

A residual formulation is employed to maintain low-frequency and local information:

$$F_DF = F_P + \gamma F_D,$$

where γ can be a learnable scalar or vector. This module serves two main purposes. First, it enables distant but structurally related areas to communicate. Second, it dynamically adjusts the importance of different feature patterns according to image content.

3.2.4 Residual-purpose U-shaped neural reconstruction module

The dynamically fused representation is subsequently supplied to a residual-purpose neural module based on an encoder-decoder U-shaped architecture.

Before entering the U-shaped module, the feature representation is aligned with the target spatial dimensions:

$$F_U = I_4(F_DF),$$

where I_4 indicates the factor-four spatial interpolation or learned upsampling specified for the network.

The encoder progressively extracts features at different semantic levels:

$$E_1 = C_1(F_U),$$

$$E_2 = C_2(D(E_1)),$$

$$E_3 = C_3(D(E_2)),$$

where C_i denotes convolutional processing and D denotes spatial downsampling.

The decoder progressively reconstructs spatial resolution:

$$D_2 = C'_2(\text{Concat}[U(E_3), E_2]),$$

$$D_1 = C'_1(\text{Concat}[U(D_2), E_1]).$$

Skip connections transfer high-frequency details directly from the encoder to the corresponding decoder levels, reducing information loss during downsampling.

Instead of forcing the network to generate the complete HRMS image, the module learns a residual:

$$R_MS = R_U(F_DF),$$

and the reconstructed high-resolution multispectral result is

$$M_HR_hat = M^{up} + R_MS.$$

This global residual formulation has a clear interpretation: M^{up} already contains the primary spectral content, while the network estimates the missing information necessary for improved high-resolution reconstruction.

3.2.5 Spectral-reconstruction objective

For reduced-resolution supervised training, the multispectral reconstruction network can be optimized with

$$L_{\text{MS}} = \lambda_1 L_1 + \lambda_{\text{SSIM}} L_{\text{SSIM}} + \lambda_{\text{SAM}} L_{\text{SAM}}.$$

The pixel reconstruction term is

$$L_1 = (1/N) \| M_{\text{HR_hat}} - M_{\text{GT}} \|_1.$$

The structural term is

$$L_{\text{SSIM}} = 1 - \text{SSIM}(M_{\text{HR_hat}}, M_{\text{GT}}),$$

while the spectral-angle loss is

$$L_{\text{SAM}} = (1/N) \sum_i \arccos((m_{\text{hat}_i}^T m_i) / (\|m_{\text{hat}_i}\|_2 \|m_i\|_2 + \epsilon)).$$

This combined objective simultaneously encourages radiometric similarity, structural preservation, and spectral-direction consistency.

3.3 Spatial information-enhanced panchromatic imaging

The second major component independently improves the spatial-feature representation of the remotely sensed PAN image. The central design is a two-channel neural network comprising a three-layer basic convolution branch for shallow spatial information; a multiscale feature-extraction branch for deeper contextual information; feature integration; four residual blocks; and image-level residual addition to the original panchromatic image.

Let the original PAN image be P .

3.3.1 Shallow three-layer convolutional channel

The first channel extracts local edges, brightness transitions, narrow structures, and fine textures:

$$S_1 = \phi(W_1^s * P + b_1^s),$$

$$S_2 = \phi(W_2^s * S_1 + b_2^s),$$

$$F_{\text{shallow}} = \phi(W_3^s * S_2 + b_3^s).$$

Because early convolutional layers preserve detailed local structures, this branch specializes in shallow information.

3.3.2 Deep multiscale channel

The second channel uses multiscale receptive fields:

$$F_{\text{deep}} = \text{MFEM_PAN}(P).$$

As with the multispectral MFEM, kernels or dilations of different sizes provide responses to edges, objects, and contextual structures at different scales. The two branches are then integrated:

$$F_{\text{TC}} = \text{Concat}[F_{\text{shallow}}, F_{\text{deep}}],$$

followed by channel projection:

$$F_0 = W_{(1 \times 1)}^p * F_{\text{TC}}.$$

3.3.3 Four-block residual refinement

Four residual blocks refine the fused shallow-deep representation:

$$F_k = F_{(k-1)} + R_k(F_{(k-1)}), \quad k = 1, 2, 3, 4.$$

Each block may contain two convolutional layers and nonlinear activation:

$$R_k(X) = W_{(k,2)} * \phi(W_{(k,1)} * X).$$

After four blocks, a reconstruction layer predicts the spatial-detail residual:

$$R_{\text{PAN}} = W_{\text{out}} * F_4.$$

The final enhanced PAN image is

$$P_{\text{hat}} = P + R_{\text{PAN}}.$$

The global residual connection is particularly important. Instead of generating the entire PAN image from latent features, the network estimates only the additional spatial detail. This preserves the basic radiometric content of the original PAN image and facilitates stable optimization.

3.4 Fusion architecture for low-resolution multispectral and high-resolution panchromatic images

The final stage integrates the two enhanced representations.

3.4.1 High-resolution multispectral reconstruction

The original LRMS image is processed through the multiscale dynamic spectral network:

$$M_{\text{HR_hat}} = S_{\text{MS}}(M).$$

For the RGB bands,

$$M_{\text{RGB_hat}} = [R_{\text{hat}}, G_{\text{hat}}, B_{\text{hat}}].$$

3.4.2 RGB-to-IHS transformation

The reconstructed RGB image is converted into the IHS color space:

$$[I, H, S] = T_{\text{IHS}}(M_{\text{RGB_hat}}).$$

A simple intensity definition is

$$I = (R_{\text{hat}} + G_{\text{hat}} + B_{\text{hat}}) / 3.$$

The hue and saturation components encode chromatic properties and should be changed as little as possible during PAN-detail incorporation.

3.4.3 Panchromatic enhancement

In parallel,

$$P_{\text{hat}} = E_{\text{PAN}}(P).$$

Direct replacement of I by P_{hat} may lead to spectral distortion because of different radiometric distributions. Therefore, the enhanced PAN component is first normalized:

$$P_{\text{tilde}} = [\sigma_I / (\sigma_{P_{\text{hat}}} + \epsilon)] (P_{\text{hat}} - \mu_{P_{\text{hat}}}) + \mu_I,$$

where μ and σ represent image mean and standard deviation.

3.4.4 Intensity fusion

The most direct IHS substitution is

$$I_{\text{F}} = P_{\text{tilde}}.$$

For improved control over spectral distortion, a weighted formulation can instead be used:

$$I_{\text{F}} = (1 - \alpha) I + \alpha P_{\text{tilde}}, \quad 0 \leq \alpha \leq 1.$$

Alternatively, only PAN high-frequency details can be injected:

$$I_{\text{F}} = I + \beta [P_{\text{tilde}} - L(P_{\text{tilde}})],$$

where L is a low-pass filter and β controls injection strength. The final fused image becomes

$$F_{\text{RGB}} = T_{\text{IHS}}^{-1}(I_{\text{F}}, H, S).$$

Therefore, the high-resolution spectral reconstruction supplies the color and spectral basis, whereas the enhanced PAN branch supplies fine spatial information.

3.5 Overall optimization strategy

The proposed system may be optimized sequentially or jointly. In sequential optimization, the multispectral reconstruction network is first trained, the PAN spatial-enhancement network is then trained, and radiometric IHS fusion is subsequently applied.

For end-to-end fine-tuning, a total objective can be defined as

$$L_{\text{total}} = \lambda_{\text{rec}} L_{\text{rec}} + \lambda_{\text{spec}} L_{\text{spec}} + \lambda_{\text{spat}} L_{\text{spat}} + \lambda_{\text{grad}} L_{\text{grad}}.$$

The reconstruction term measures direct output error:

$$L_{\text{rec}} = \|F - M_{\text{GT}}\|_1.$$

Spectral consistency can be enforced by degrading the fused output to LRMS resolution:

$$L_{\text{spec}} = \|D_{\text{MS}}(F) - M\|_1.$$

Spatial consistency can compare the intensity or PAN-like projection of the output with the PAN observation:

$$L_{\text{spat}} = \|H(F) - P\|_1.$$

An edge loss is

$$L_{\text{grad}} = \|\text{grad}(F) - \text{grad}(M_{\text{GT}})\|_1.$$

This combination prevents optimization from concentrating exclusively on pixelwise reconstruction while neglecting spectral fidelity or edge quality.

4. Experimental Results and Analysis

4.1 Important note on experimental evidence

The provided design material does not contain actual implementation outputs, dataset partitions, training logs, quantitative evaluation scores, or visual comparison figures. Consequently, no scientifically defensible paper can claim specific improvements such as "2.3 dB higher PSNR" or "5% lower SAM" without running the proposed model. Accordingly, this section provides the complete experiment design and publication-ready result structure. All fields marked [Measured value] must be replaced by actual measurements.

4.2 Recommended datasets

A comprehensive evaluation should include multiple satellite sensors with different spectral responses and spatial resolutions. WorldView-2, WorldView-3, QuickBird, and GaoFen-2 are widely employed in contemporary pansharpening research and allow evaluation of both reduced-resolution reconstruction and full-resolution generalization.

Dataset	PAN resolution	MS resolution	Bands used	Training patches	Test patches
WorldView-3	[Insert]	[Insert]	RGB / multispectral	[Insert]	[Insert]
WorldView-2	[Insert]	[Insert]	RGB / multispectral	[Insert]	[Insert]
QuickBird	[Insert]	[Insert]	RGB / multispectral	[Insert]	[Insert]
GaoFen-2	[Insert]	[Insert]	RGB / multispectral	[Insert]	[Insert]

The data should be divided into training, validation, and test sets without spatial overlap to reduce the risk of information leakage.

4.3 Reduced-resolution protocol

Because a genuine HRMS reference normally does not exist at PAN resolution, a reduced-resolution evaluation should follow a consistent degradation process. The original multispectral image serves as ground truth:

$$M_{\text{GT}} = M_{\text{original}}.$$

A simulated LRMS image is obtained through sensor-consistent low-pass filtering and downsampling:

$$M_{\text{LR}} = \text{down}_r[K_{\text{MS}} * M_{\text{GT}}].$$

Similarly, the PAN image is filtered and downsampled:

$$P_{\text{LR}} = \text{down}_r[K_{\text{PAN}} * P].$$

The network receives $(M_{\text{LR}}, P_{\text{LR}})$, and its output is directly compared with M_{GT} .

4.4 Full-resolution protocol

At native sensor resolution, no true HRMS reference is available. Therefore, full-resolution evaluation should employ no-reference quality measures that separately evaluate spectral and spatial consistency. Recommended indicators include D_lambda, D_S, and QNR, where lower D_lambda and D_S are preferable and higher QNR is preferable.

Visual inspection should additionally examine building boundaries, road structures, vegetation textures, homogeneous surfaces, color consistency, ringing artifacts, checkerboard artifacts, local oversharpening, and object-edge duplication resulting from misregistration.

4.5 Implementation details

A reproducible implementation may use PyTorch with CUDA acceleration. A suitable initial configuration is:

Parameter	Recommended setting
Upscaling factor	4
Optimizer	Adam or AdamW
Initial learning rate	1×10^{-4}
Batch size	8-32 depending on memory
Training epochs	200-500
Patch size	32 x 32 LRMS / 128 x 128 PAN for $r = 4$
Activation	ReLU, LeakyReLU, or GELU
LR scheduler	Cosine decay or step decay
Weight decay	10^{-6} to 10^{-4}
Data augmentation	Rotations and horizontal/vertical flipping

For the final manuscript, these values should be replaced by the exact configuration actually used.

The experimental hardware statement should be written in the following form: All networks were implemented using [PyTorch version] and trained on [GPU model] with [GPU memory] GB memory. The processor was [CPU model], and the system contained [RAM] GB system memory. Training required approximately [Measured value] hours for [Measured epochs] epochs.

4.6 Evaluation metrics

A robust reduced-resolution assessment should include PSNR, SSIM, SAM, ERGAS, and a multispectral quality index such as $Q2^n$.

Peak signal-to-noise ratio

$$\text{PSNR} = 10 \log_{10}(\text{MAX}^2 / \text{MSE}).$$

Higher PSNR indicates lower reconstruction error.

Structural similarity index

SSIM evaluates luminance, contrast, and structural correspondence. Higher values indicate greater structural similarity.

Spectral angle mapper

$$\text{SAM} = (1/N) \sum_i \arccos(f_i^T g_i / (\|f_i\|_2 \|g_i\|_2)).$$

Lower SAM indicates better spectral consistency.

ERGAS

ERGAS measures normalized global radiometric error. Lower values are preferable.

4.7 Comparison methods

The proposed MDFTC-Fusion method should be compared against representative methods from several methodological families: PNN [2], PanNet [3], MSDCNN [4], DiCNN [5], TFNet [6], FusionNet [7], Vision Transformer for Pansharpening [11], SDRCNN [15], CANNet [22], CMT [23], and PanAdapter [27]. This comparison provides meaningful coverage of shallow CNN, high-pass residual, multiscale, detail-injection, two-stream, transformer, dense-residual, adaptive-convolution, and pretrained-model approaches.

Table 1. Reduced-resolution quantitative comparison

Method	PSNR up	SSIM up	SAM down	ERGAS down	Q2^n up
PNN	[Measured]	[Measured]	[Measured]	[Measured]	[Measured]
PanNet	[Measured]	[Measured]	[Measured]	[Measured]	[Measured]
MSDCNN	[Measured]	[Measured]	[Measured]	[Measured]	[Measured]
DiCNN	[Measured]	[Measured]	[Measured]	[Measured]	[Measured]
TFNet	[Measured]	[Measured]	[Measured]	[Measured]	[Measured]
FusionNet	[Measured]	[Measured]	[Measured]	[Measured]	[Measured]
ViT	[Measured]	[Measured]	[Measured]	[Measured]	[Measured]
SDRCNN	[Measured]	[Measured]	[Measured]	[Measured]	[Measured]
CANNet	[Measured]	[Measured]	[Measured]	[Measured]	[Measured]
MDFTC-Fusion	[Measured]	[Measured]	[Measured]	[Measured]	[Measured]

After actual scores are obtained, the results can be described as follows: As presented in Table 1, MDFTC-Fusion achieved a PSNR of [value] dB, SSIM of [value], SAM of [value], ERGAS of [value], and Q2^n of [value]. Compared with the strongest baseline, the proposed method improved [metric] by [measured difference]. The lower SAM and ERGAS indicate improved preservation of spectral characteristics, whereas the higher PSNR, SSIM, and Q2^n demonstrate superior reconstruction accuracy and spatial-structural consistency. The results suggest that separately reconstructing the LRMS image and enhancing PAN information before controlled intensity-domain integration can mitigate the spatial-spectral interference associated with direct multimodal concatenation.

4.8 Full-resolution performance

Table 2. Full-resolution no-reference comparison

Method	D_lambda down	D_S down	QNR up
PNN	[Measured]	[Measured]	[Measured]
PanNet	[Measured]	[Measured]	[Measured]
FusionNet	[Measured]	[Measured]	[Measured]
SDRCNN	[Measured]	[Measured]	[Measured]
CANNet	[Measured]	[Measured]	[Measured]
MDFTC-Fusion	[Measured]	[Measured]	[Measured]

The corresponding discussion should explicitly avoid drawing conclusions unless the actual numbers support them. A suitable final paragraph is: At full resolution, MDFTC-Fusion obtained D_lambda = [value], D_S = [value], and QNR = [value]. The value of D_lambda demonstrates [strong/moderate/weak, according to actual result] spectral consistency, whereas D_S indicates [interpretation based on actual result] spatial agreement with the PAN reference. The resulting QNR of [value] is [higher/lower/comparable] to that of [baseline name].

4.9 Visual performance and analysis

Visual comparisons should include enlarged image crops from challenging regions. In urban regions, attention should be given to roof edges, road intersections, narrow linear structures, and high-contrast building

boundaries. In vegetation and agricultural areas, evaluation should focus on natural color retention, texture continuity, avoidance of artificial sharpening, and preservation of homogeneous spectral regions.

The anticipated architectural behavior can be explained as follows. The MFEM processes structures over multiple receptive fields and therefore has the capacity to distinguish fine local details from larger contextual patterns. The DFFM dynamically controls feature interactions rather than assigning equal significance to all scales. The U-shaped reconstruction network combines shallow information with deeper contextual features through skip connections. In the PAN branch, the shallow convolutional channel preserves edges and local textures, whereas the multiscale channel provides deeper contextual information. Four residual blocks then refine the joint representation before addition to the original PAN image.

A proper SCI manuscript should present side-by-side images for the original LRMS, PAN, each comparison method, MDFTC-Fusion, and ground truth in the reduced-resolution experiment. Error maps are also recommended.

4.10 Ablation study

The ablation analysis should demonstrate the independent contribution of each component. The recommended variants are:

- Variant A: Remove the multiscale feature extraction module.
- Variant B: Remove dynamic self-attention and use simple concatenation.
- Variant C: Replace the residual U-shaped reconstruction module with ordinary convolution.
- Variant D: Remove the shallow branch from the two-channel PAN network.
- Variant E: Remove the deep multiscale PAN branch.
- Variant F: Remove the four residual refinement blocks.
- Variant G: Use the original PAN image instead of the enhanced PAN image.
- Variant H: Replace radiometrically controlled IHS fusion with direct PAN substitution.
- Full model: Complete MDFTC-Fusion.

Table 3. Component ablation

Configuration	MFEM	DFFM	U-Net reconstruction	Two-channel PAN	Residual blocks	PSNR up	SAM down
A	No	Yes	Yes	Yes	Yes	[Measured]	[Measured]
B	Yes	No	Yes	Yes	Yes	[Measured]	[Measured]
C	Yes	Yes	No	Yes	Yes	[Measured]	[Measured]
D	Yes	Yes	Yes	Deep only	Yes	[Measured]	[Measured]
E	Yes	Yes	Yes	Shallow only	Yes	[Measured]	[Measured]
F	Yes	Yes	Yes	Yes	No	[Measured]	[Measured]
Full model	Yes	Yes	Yes	Yes	Yes	[Measured]	[Measured]

4.10.1 Effect of multiscale feature extraction

The MFEM is designed to improve representation of heterogeneous remote sensing structures by using different receptive fields. Its contribution should be demonstrated by comparing the complete method against Variant A. Improvements in PSNR and SSIM would indicate better reconstruction accuracy, while a reduction in SAM would suggest that the multiscale processing does not compromise spectral fidelity.

4.10.2 Effect of dynamic feature fusion

Variant B replaces self-attention-based dynamic integration with fixed feature combination. The comparison is intended to determine whether input-dependent weighting provides an advantage over static fusion. After measurement, a rigorous interpretation should read: Removing DFFM changed PSNR from [full value] to [ablation value] and SAM from [full value] to [ablation value], demonstrating [the conclusion actually justified by the data].

4.10.3 Effect of the residual U-shaped network

The U-shaped module combines global semantic information and local spatial detail. Skip connections are expected to reduce information loss introduced by encoder downsampling. Variant C determines whether this multilevel reconstruction mechanism offers measurable improvement over ordinary convolutional reconstruction.

4.10.4 Effect of two-channel PAN feature extraction

The two PAN channels have intentionally different purposes. The shallow branch emphasizes local detail, whereas the multiscale branch extracts deeper structural patterns. Variants D and E should determine whether their combination is more effective than either channel independently.

4.10.5 Effect of PAN enhancement

Variant G directly uses the original PAN image in the IHS fusion process. The full model instead uses

$$\hat{P} = P + R_{\text{PAN}}.$$

This experiment directly measures whether the two-channel PAN enhancement network contributes useful spatial information.

4.10.6 Effect of radiometric matching

Direct replacement of the IHS intensity component can cause color distortion when the statistical distribution of PAN differs from that of the multispectral intensity. Variant H therefore evaluates the effect of removing radiometric matching.

4.11 Computational complexity

SCI reviewers increasingly expect information about computational cost. The following values should therefore be measured:

Method	Parameters (M)	FLOPs (G)	Inference time (ms)
PNN	[Measured]	[Measured]	[Measured]
PanNet	[Measured]	[Measured]	[Measured]
FusionNet	[Measured]	[Measured]	[Measured]
CANNet	[Measured]	[Measured]	[Measured]
MDFTC-Fusion	[Measured]	[Measured]	[Measured]

The discussion should explicitly recognize any accuracy-complexity trade-off. A method that improves quality but requires much higher computational cost should not be described as universally superior without qualification.

5. Conclusion

This paper proposed MDFTC-Fusion, a multistage deep-learning framework for fusing low-spatial-resolution multispectral imagery with high-spatial-resolution panchromatic imagery. Unlike conventional end-to-end schemes that directly mix the two modalities at an early processing stage, the proposed approach decomposes the pansharpening problem into three explicit subtasks: high-resolution multispectral reconstruction, panchromatic spatial enhancement, and controlled IHS-domain fusion.

For multispectral reconstruction, a multiscale feature dynamic interaction network was introduced. The network combines a Multi-Scale Feature Extraction Module for obtaining representations with different receptive fields, a Dynamic Feature Fusion Module based on self-attention for adaptive integration of those representations, and a residual-purpose U-shaped encoder-decoder for reconstruction of high-resolution spectral imagery. This design is intended to enhance both local detail and global contextual understanding while retaining information through residual and skip connections.

For panchromatic enhancement, a two-channel neural network was constructed. A three-layer basic convolutional branch extracts shallow spatial details, whereas a parallel multiscale branch captures deeper information. The resulting features are integrated and successively refined by four residual blocks. The predicted spatial residual is added directly to the original PAN observation to preserve fundamental radiometric information.

Finally, the high-resolution RGB reconstruction is transformed into the IHS color space. The enhanced PAN image is radiometrically adjusted and incorporated into the intensity component, after which inverse transformation produces the fused high-resolution output. The separation of the spectral reconstruction and spatial-enhancement tasks provides a clearer and more interpretable mechanism for controlling the classical trade-off between spatial sharpness and spectral fidelity.

Several limitations should nevertheless be acknowledged. First, conventional IHS fusion is naturally suited to three-band RGB imagery; extension to arbitrary multispectral band counts requires generalized intensity transformations or bandwise spatial-detail injection. Second, self-attention may increase memory consumption for large remote sensing images. Third, performance can deteriorate in the presence of significant spatial misregistration between PAN and MS observations. Fourth, generalization across satellite sensors may be affected by differences in modulation transfer functions, spectral-response characteristics, noise distributions, atmospheric conditions, and ground sampling distances.

Future research should consequently investigate sensor-aware degradation modeling, deformable or correspondence-aware alignment, linear-complexity attention, frequency-domain spatial-spectral decomposition, self-supervised full-resolution adaptation, and generalization to arbitrary multispectral and hyperspectral band numbers. Another promising direction is to replace fixed IHS intensity fusion with a learnable but physically constrained detail-injection module. Such developments could further strengthen spatial reconstruction while preserving spectral reliability and computational efficiency.

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